

Support Descent for All-Prime Primitive Orientations

Autonomous discussion Pass 105 isolates the support-descent statement for the all-prime primitive-orientation package.

For finite support S , the primitive orientations are

$$\mathcal{O}_S = \{c = (c_p)_{p \in S} \in \mathbb{Z}^S : \sum_{p \in S} c_p = 0, \gcd(c_p) = 1\}.$$

If $S \subseteq T$, the canonical orientation map is zero-extension

$$e_{S,T} : \mathcal{O}_S \rightarrow \mathcal{O}_T.$$

It preserves zero-sum, primitivity, and the antipode, and it composes strictly along support chains.

The boundary support map has the opposite variance:

$$T_T \rightarrow T_S.$$

There is no total restriction map on primitive orientations $\mathcal{O}_T \rightarrow \mathcal{O}_S$. Deleting coordinates can destroy zero-sum:

$$(1, 1, -2) \in \mathcal{O}_{\{2,3,5\}}$$

restricts to $(1, 1)$ on $\{2, 3\}$, whose sum is 2.

Therefore the all-prime primitive-orientation object is a filtered colimit by zero-padding. It may be represented as primitive finitely supported zero-sum integer functionals on the set of all primes, modulo padded zero coordinates.

The antipode quotient gives primitive lines $[c] = \{c, -c\}$, but the sign must still be retained as the $B\mathbb{Z}/2$ local system on the boundary/Yoneda line from Pass 104. The correct categorical package is therefore a span-stack or Grothendieck presentation combining:

- zero-extension on orientation torsors;
- support projection on boundary groups;
- the $B\mathbb{Z}/2$ boundary-line local system.

It is not a plain sheaf of primitive orientations on finite supports with restriction maps, and it does not create a degree-0 Weyl/Fourier morphism $\epsilon \rightarrow \mathbb{Q}$.

The checker `artifacts/reports/pass105-support-descent-primitive-orientations-check.json` verifies zero-extension preservation and functoriality, explicit restriction failure, colimit padding equivalence, nonexistence of support-symmetric primitive orientations, and the span-stack verdict.

The next task is to compute the obstruction to stackifying primitive orientations over finite supports when restriction maps are demanded, then state the universal property of the span-stack/left-Kan colimit package.